

MR 6-Channel Flexible Phased Array Coil

1.5T and 3.0T Signa[®] HD and HDx
Operator/Service Manual



MR 6-Channel Flexible Phased Array Coil
Operator Manual, English
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Revision 2
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Medical Device Directive

This product conforms with the requirements of council directive 93/42/EEC concerning medical devices and bears the following CE Conformity Marking:



European Representative:

GE Medical Systems S.C.S
Quality Assurance Manager
283 rue de la Minière
78530 BUC France
Telephone: +33 1 30 70 40 40

Do not use devices that transmit RF Signals (**cellular phones**, transceivers, or radio controlled products) in the vicinity of this equipment as they may cause performance outside the published specifications. Keep the power to these types of devices turned off when near this equipment.

The medical staff in charge of this equipment is required to instruct technicians, patients, and other people who may be around this equipment to fully comply with the above requirement.

Immunity/Emissions Exceptions: Note the exceptions from the EMC test results. Check with the business EMC engineer for this information.

In accordance with the international safety standard IEC 60601-1, this system is a Class II device, acceptable for Continuous Operation, having ordinary protection against ingress of water (IPX0) with type BF applied parts and is not for use in the presence of flammable anesthetics.





CAUTION: This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.

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Chapter 1

Safety

Introduction

This chapter describes the safety precautions of the GE Healthcare **6-Channel Flexible Phased Array coil**, compatible with the GE 1.5T and 3.0T Signa® HD and HDx whole body MRI systems with a minimum of eight receive channels. This manual contains detailed information on the set-up, positioning, and use of your coil. The instructions should be read carefully and thoroughly before attempting to scan patients with the coil.

The coil is indicated for use, on the order of a physician in conjunction with an MR scanner, as an accessory to produce 2D and 3D images. The Flexible Array coil is optimized to scan the head for brain imaging.

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to prevent patient exposure to electrical or mechanical hazards that may cause injury.

If you have any questions or comments on this manual, or need any assistance with the use of the product, please contact your GE Healthcare representative at: 1-800-GE CARES.



CAUTION: Federal law restricts this device to sale, distribution, and use by or on the order of a physician.

You should be familiar with the following safety guidelines before using the **6-Channel Flexible Phased Array coil**:

- [Use of symbols](#)
- [Indications for use](#)
- [Coil operating safety](#)
- [Patient safety](#)
- [Equipment safety](#)
- [Electrical and mechanical safety](#)
- [Incident reporting](#)

Use of symbols

The following safety notices are used to emphasize certain safety instructions. This manual uses the international symbol along with the danger, warning, or caution message.



DANGER: Danger is used to conditions or actions for which a specific hazard is known to exist which will cause severe personal injury, death, or substantial property damage if the instructions are ignored.



WARNING: Warning is used to identify conditions or actions for which a specific hazard is known to exist which may cause severe personal injury, death, or substantial property damage if the instructions are ignored.



CAUTION: Caution is used to identify conditions or actions for which a potential hazard may exist which will or can cause minor personal injury or property damage if the instructions are ignored.



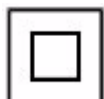
Coil CAUTION: Coil Caution is used to identify conditions or actions for which a potential hazard of crossing or looping coil cables may exist which will or can cause minor personal injury or property damage if the instructions are ignored.

IMPORTANT: Important indicates information where adherence to procedures is crucial or where your comprehension is necessary to apply a concept or effectively use the product.

NOTE: A Note provides additional information that is helpful to you. It may emphasize certain information regarding special tools or techniques, items to check before proceeding, or factors to consider about a concept or task.

One or more of the product identification labels may be on your imaging coils. Please familiarize yourself with the labels that apply to your particular peripheral equipment.

Table 1-1 Product identification labels

Label	Description (typical use)
	Class II Equipment (double insulated) (ratings)
	Type BF Applied Part
	Attention – Consult accompanying documents

Indications for use

The coil is indicated for use, on the order of a physician in conjunction with an MR scanner, as an accessory to produce 2D and 3D images.

The intended areas of interest to be scanned with the Flexible Array coil include, but are not limited to: head, knee, shoulder, foot, wrist, and other extremities.



CAUTION: This device is limited by federal law to investigational use for indications not in the “Indications for use” statement.



CAUTION: Federal law restricts the sale, distribution, and use of this device to or on the order of a physician.

Coil operating safety



WARNING: The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system operators manual.

- Cables should not be looped or crossed as arcing and patient burns could result.
- Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.
- Do not allow the surface coil cable to touch the patient; patient burns can result. Use a thermal resistant material or pad to keep the cable from touching the patient.
- Keep the length of the cable in the bore to a minimum. Avoid bending the cable 180 degrees and route the cables out of the bore in the most direct way - without looping or coiling.
- Visually inspect the cable insulator jackets, strain reliefs and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.
- Remove unplugged surface coils or unused accessory devices from the magnet bore; a patient burn can result.
- Remove blue wire-type ECG gating cables from the magnet bore before scanning with surface coils.
- Do not use blue wire-type ECG leads when you are doing gated surface coil scans. When they are used in the presence of a surface coil cable, the potential for heating increases. Use peripheral gating or high impedance ECG leads rather than blue wire-type ECG leads when you are doing gated surface coil scans.
- When using any coil, make sure the patient's body does not touch the magnet bore. Use thermal resistant material or padding between the patient and the magnet to prevent burns that could be caused by patient contact with the interior of the magnet bore.
- Use only GE or GE-authorized accessory coils and cables.
- Use only accessories in good condition. If you suspect that an accessory is not in good condition, discontinue its use and contact your local GE service engineer.
- Do not use surface coils with exposed metal conductors or damaged insulation. Skin contact with metal conductors can cause burns.

Quality assurance

The procedure described in the Quality assurance chapter of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

Coil modifications

Any changes or modifications to the coil must be approved and performed by GE Healthcare prior to installation.

Patient safety



WARNING: The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system operators manual.

- If a patient complains of burning, tingling, stinging, or other similar "burn"-type sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.
- Ensure that the patient is comfortably positioned.
- Monitor the patient periodically. Stop the scan immediately if the patient reports any heating, burning or tingling sensations.



Coil CAUTION: Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Emergency procedures

In the unlikely event that a coil creates smoke, sparks, or makes an unusually loud noise, or if the patient requires emergency assistance:

- stop the scan if one is in progress
- remove the patient from the scan room if medical treatment is needed

Patient contraindications



WARNING: Contraindications

- Patients with ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.
- Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.
- There is a risk to scanning feverish or decompensated cardiac patients.
- Facial makeup should be removed before scanning because it may contain metal flakes, which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.
- Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing an MR exam.

Equipment safety

All personnel using the system should be instructed in the proper operation and use of the system. Personnel should observe all warnings and cautions that appear in the manual.



WARNING: Auxiliary equipment (such as gating equipment, vital signs monitoring systems and RF coils) that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.



WARNING: The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen or nitrous oxide.



Coil CAUTION: Prevent cables from forming loops. Looping degrades scan performance of the coil through RF coupling.



WARNING: This equipment is not protected against ingress of water (IPX0).

NOTE: This equipment is rated for continuous operation.

NOTE: For the disposal of this coil at the end of its useful life, please contact the GE Service Personnel.

Electrical and mechanical safety



WARNING: The Coil consists of both electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it for any external damage. Do not use the coil if the housing or cable is broken.

Incident reporting

Contact GE Healthcare immediately to report an incident and/or injury to a patient, operator, or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

For additional information, contact:

GE Healthcare – Americas

- USA: 800-558-5102
- Canada: 800-668-0732

GE Healthcare – Asia

- India: 91-80-845-2923
- Japan: 81-120-48-2630
- China: 86-21-621-92228
- Australia: 61-2-9975-5501
- Singapore: 65-291-8528
- Taiwan: 886-2-2505-7900

GE Healthcare – Europe: 33-1-41-19-7676

GEMS on the World Wide Web: <http://www.gemedicalsystems.com/rad/mri/index.html>

GE Healthcare
3200 N. Grandview Blvd.
Waukesha, WI 53188
USA

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Chapter 2

MR Flexible Phased Array coil

Introduction

The 1.5T and 3.0T **6-Channel Flexible Phased Array coil**, compatible with the GE 1.5T and 3.0T Signa® HD and HDx whole body MRI systems, is a receive only, multi-coil array designed to give the optimal signal-to-noise ratio (SNR) and uniformity for high resolution head examinations. The coil enables you to image multi-oblique slices of the head. The multi-element design improves the coverage and resolution. All compatible systems require a minimum of eight receive channels.

When using the MR system for imaging using the **6-Channel Flexible Phased Array coil**, refer to this manual for coil details while using existing product documentation for details of diagnostic scanning of patient.

The **6-Channel Flexible Phased Array coil** topics discussed in within this chapter include:

- [Shipping contents](#)
- [Damage in transportation](#)
- [Product Locator](#)
- [Requirements and specifications](#)
- [Coil description](#)
- [Coil positioning](#)
- [Patient positioning](#)
- [Modes of operation](#)

Shipping contents

The following tables include the coil contents shipped with the specified catalog number.

Table 2-1 Catalog M0002SS shipping contents

Description	Part number	Quantity
1.5T 6-Channel Flex Phased Array coil assembly with a Legacy Intermediate Adaptor Connector	5229622	1
Operator/Service manual	5267963	1

Table 2-2 Catalog M0003SS shipping contents

Description	Part number	Quantity
1.5T 6-Channel Flex Phased Array coil assembly with a Hypertronic Intermediate Adaptor Connector	5229625	1
Operator/Service manual	5267963	1

Table 2-3 Catalog M0004SS shipping contents

Description	Part number	Quantity
3.0T 6-Channel Flex Phased Array coil assembly with a Legacy Intermediate Adaptor Connector	5229645	1
Operator/Service manual	5267963	1

Table 2-4 Catalog M0005SS shipping contents

Description	Part number	Quantity
3.0T 6-Channel Flex Phased Array coil assembly with a Hypertronic Intermediate Adaptor Connector	5229649	1
Operator/Service manual	5267963	1

Damage in transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366
- Contact your local service coordinator for more information on this process.

Product Locator

The Global Install Base Database (also known as the Product Locator system) tracks shipment, trans-shipment, and field location of the serialized models. There are two methods for submitting Product Locator information.

Method 1: At this time, for U.S. ONLY, the preferred method of submitting information is the FE Site Verification web site at: <http://gein2.med.ge.com/gib>.

- The FE Site Verification has three components that are available on the web from the main menu.
 - Install/deinstall product locator model and serial numbers
 - Add/modify ship to address information
 - Update CARES FE data for primary/secondary FEs
- To obtain a copy of the FE training tutorial for using this web site, a downloadable copy is available at the Product Locator Support Central Page: <http://supportcentral.ge.com/15563>.

Method 2: One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

- Verify that serial and model number on each rating plate matches the installation card numbers before removing installation card.
- Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

IMPORTANT: The 6-Channel Flex Coil catalog contains three different serialized components. Make sure to submit three Installation Cards for this catalog.

Requirements and specifications

Coil installation and configuration requires the services of your GE Service Representative. Store the coil in an air-conditioned scan room or equipment room.

Table 2-5 Environmental requirements

Measurement type	Transport and storage	Operating conditions
Barometric pressure	500 to 1060 hPa	700 to 1060 hPa
Humidity	10 to 100% non-condensing	10 to 90% non-condensing
Temperature	-40 to 65° C (-40 to 149° F)	15 to 21° C (59 to 69.8° F)

Table 2-6 Coil frequency specifications

Field strength	Coil model	Frequency
1.5T	6-Channel Flex Phased Array coil	63.86 MHz ± 50 KHz
3.0T	6-Channel Flex Phased Array coil	127.73 MHz ± 50 KHz

Coil description

The 1.5T and 3.0T **6-Channel Flexible Phased Array** is a receive only, multi-coil array optimized for high resolution head examinations. The coil enables you to image multi-oblique slices of the head. The coil is indicated for use, on the order of a physician in conjunction with an MR scanner, as an accessory to produce 2D and 3D images.

The **6-Channel Flexible Phased Array coil** consists of two flexible parts enabling large opening and flexibility. Both halves are flexible in order to conform to the contours of the patient, and to accommodate the patient's anatomy. This maximizes signal integrity and clinical utility. Both coil's halves are connected to the system through an intermediate connector, which connects directly to the Low Profile Carriage Assembly (LPCA) of the MRI system. Coil setup flexibility allows the use of only one coil half if desired.



CAUTION: Correctly aligning the array provides a uniform signal across the sensitive volume, eliminating bright spots, high intensity areas and signal drop-off. Each of these halves has a cable emerging out to the point where they plug into an intermediate connector, which is then connected to the LPCA of the MR system.

Figure 2-1 6-Channel Flexible Phased Array coil with Legacy and Hypertronics connectors



Coil positioning

To position the **6-Channel Flexible Phased Array coil**, perform the following procedures for routine scanning.



CAUTION: Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from coupling.

Routine coil positioning procedures

1. Place one half of the coil on the patient cradle so that it sits snugly as shown in Figure 2-2.
2. Position the patient so that there is no direct contact between the patient's skin and magnet bore or coil.
 - Use non-conductive padding that is at least 0.25 inches thick between the patient's skin and the magnet bore.
 - Use additional pads to immobilize the patient and make them comfortable.
 - Place a clean cotton sheet over the coil and comfort pad so the patient's skin does not come in contact with the coil or the comfort pad.
3. Place the coil around the patient's head.

Figure 2-2 Coil setup



4. Place the coil in a way such that the desired region of the patient will be centered on the coil.



For instructions on coil positioning procedures with the Surgical Suite, please refer to the Surgical Suite operator manual, publication number 5310534. For coil positioning instructions with Oncology procedures, please refer to the Oncology Package, publication number 5220549.

Patient positioning

To position the patient, perform the following steps:

1. Position the patient on the posterior half of the array coil, between the two coil halves.
 - If indicated, prepare the patient with the **Respiratory Bellows** and/or **ECG leads** before proceeding.
2. With the patient lying down, gently use pads or straps as necessary to position the coil in the desired position for imaging (Figure 2-3).

Figure 2-3 Patient setup with a 6-channel Flex Coil



3. Provide earplugs for the patient after all instructions have been given.



WARNING: Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

- Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better (e.g., 30 dB, 32 dB, etc.).
4. Plug the **Coil Connectors** into the **Intermediate connector**.
 5. Connect the **Intermediate connector** into the appropriate **Phased Array Port** on the LPCA at the head end of the patient cradle (Figure 2-4, Figure 2-5, or Figure 2-6).

Figure 2-4 1.5T Legacy Phased Array Port C: Legacy connector (1) and Hypertronic connectors (2)

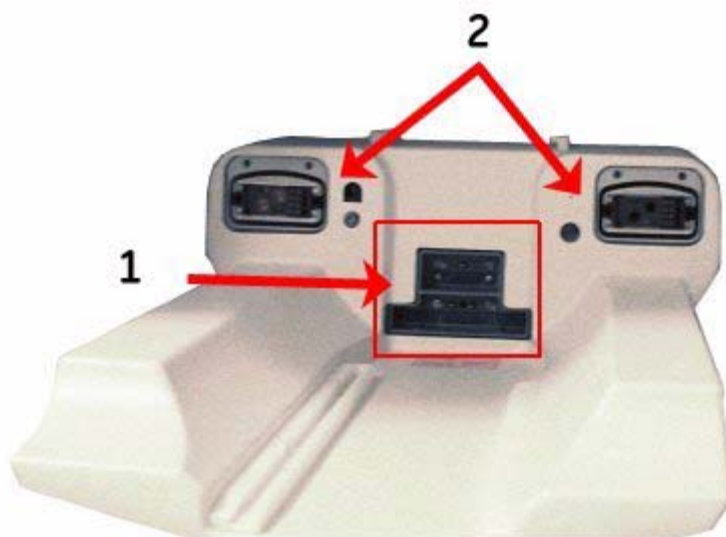


Figure 2-5 3.0T Legacy Phased Array Port C

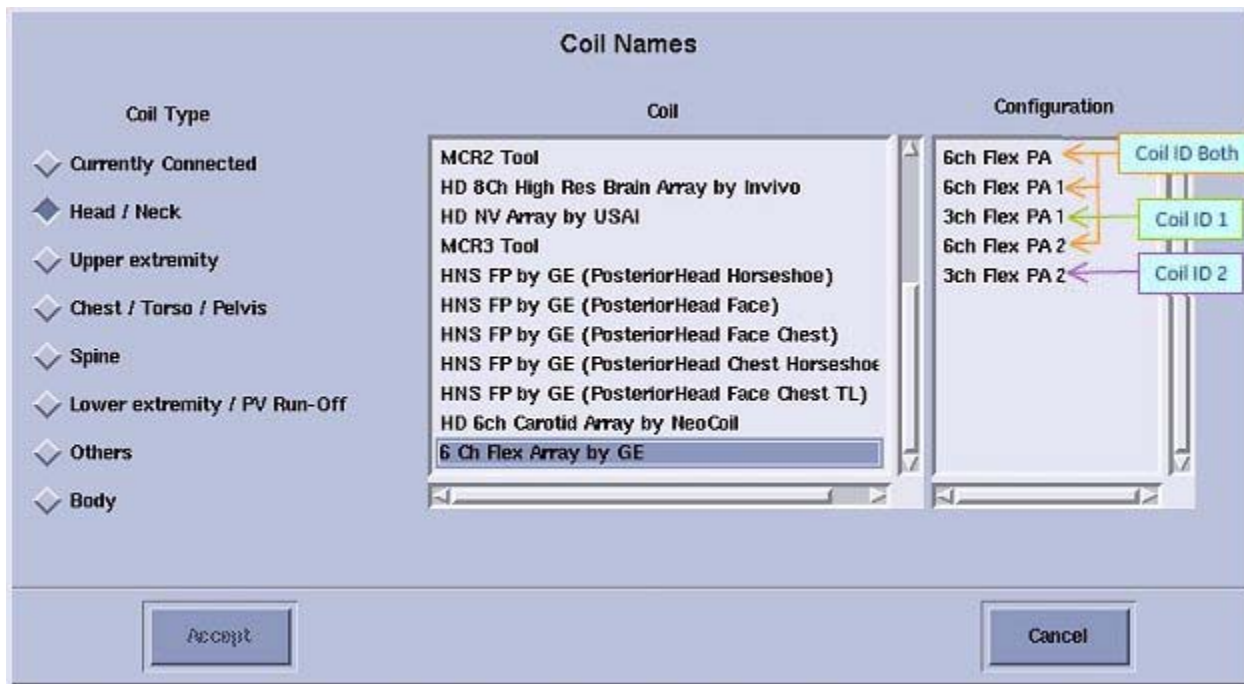


Figure 2-6 1.5T/3.0T HDx Port A with System Preamps



6. Make sure that the coil cable is not looping at any point within the magnet bore.
7. Proceed with scan prescription and select the **6 Ch Flex Array by GE coil** at the coil screen.
 - For HDx upgraded systems, the coil description is **6 Ch Flex Array Upg by GE**.
 - With both halves of the coil connected, you have full flexibility to choose the whole coil or one half configuration.
 - The Currently Connected feature on HDx systems displays the connected coil name and available configuration.

Figure 2-7 HDx Coil Names screen



8. Instruct the patient to close his or her eyes, and then turn the Laser Alignment Light on.
 - The Laser Alignment Light for patient positioning can cause eye injury.
9. Align the laser or localizing light so that the desired head region lies at the center of the coil.
 - To aid in the centering of the coil in the magnet, positioning aids have been provided on the coil in the form of dark cross hairs.
 - The localizing light should be aligned with these cross hairs for positioning accuracy within the bore.



CAUTION: Instruct the patient to close his or her eyes when landmarking and when the laser light is turned on. Exposing eyes to the laser alignment lights may result in eye injury.

10. Use the system recommended protocols for scanning with this coil for optimal results and best performance.
 - Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

Modes of operation

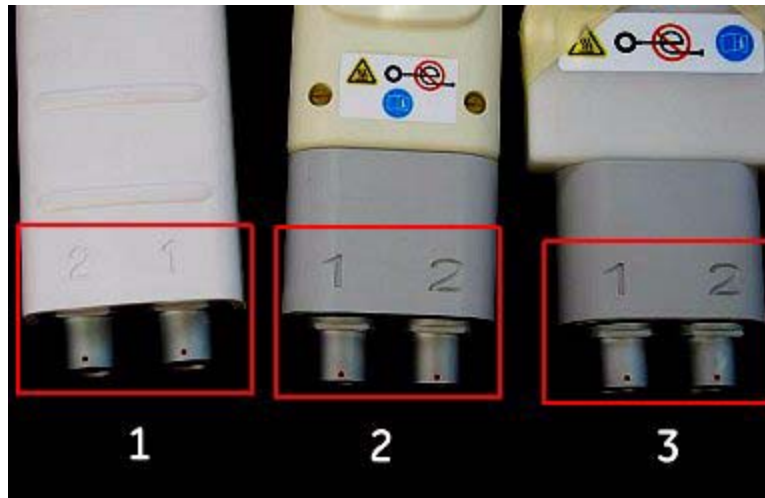
The **6-channel Flexible Phased Array coil** can be operated in three modes for the Legacy version and five modes for the Hypertronics version. See Figure 2-8 for an understanding of the connector versions.

- **6ch Flex PA:** in this mode, both coil halves are connected to the Intermediate connector. Reconstruction data is taken from all six channels.
- **6ch Flex PA 1:** in this mode, both coil halves are connected to the Intermediate connector, but only three channels of the coil connected to port 1 are reconstructed (one half of the coil is active).
 - For 1.5T HD and upgraded 1.5T HDx, this mode can be used when only one half of the coil is connected to port 1. See [Figure 2-9](#), if the coil is plugged into port 2 and you select this mode, a system error occurs.
- **6ch Flex PA 2:** in this mode, both coil halves are connected to the Intermediate connector, but only three channels of the coil connected to port 2 are reconstructed (one half of the coil is active).
 - For 1.5T HD and upgraded 1.5T HDx, this mode can be used when only one half of the coil is connected to port 2. See [Figure 2-9](#), if the coil is plugged into port 1 and you select this mode, a system error occurs.
- **3ch Flex PA 1:** in this mode, only one half of the coil is connected to port 1 and three channels are reconstructed.
 - This mode is only applicable for 1.5T and 3.0T systems with the Hypertronics connector.
- **3ch Flex PA 2:** in this mode, only one half of the coil is connected to port 2 and three channels are reconstructed.
 - This mode is only applicable for 1.5T and 3.0T systems with the Hypertronics connector.

Figure 2-8 6-Channel Flex Coil connectors: 1.5T/3.0T Hypertronics (1) , 3.0T Legacy (2), and 1.5T Legacy (3)



Figure 2-9 Connector port numbers: 1.5T/3.0T Hypertronics (1), 3.0T Legacy (2), and 1.5T Legacy (3)



Chapter 3

Quality assurance

Introduction

Quality control is an important part of the 1.5T and 3.0T **6-Channel Flexible Phased Array coil**. A quality assurance test allows you to establish a baseline of coil performance before you begin patient scanning and check for maintained quality performance throughout the coil's use.

The **6-Channel Flexible Phased Array coil** quality assurance components discussed in this chapter include:

- [Functional check](#)
- [Performance check](#)
- [SNR image analysis](#)

NOTE: The coil quality assurance test should take one hour or less to complete.

Functional check

The scanner verification and coil configuration functional checks should be performed by your service representative.

Scanner verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

Coil configuration

Before using the coil, check whether the coil configuration for the 6-Channel Flex Coil has been loaded onto the scanner.

1. From the Service Desktop, select **Configuration > Config File Manager > Click Here to Start this Tool**.
2. On the left hand side of the screen, click **Coil Config**.
3. Click the **Site Coils** menu and look for the appropriate 6ch Flex coil configurations. See Table 3-1 for coil names and compatibility.

Table 3-1 Coil names and system compatibility

Coil name	1.5T HD LPCA Legacy Port 12x sw	Upgrade to 1.5T HDx LPCA Legacy Port 14x sw	1.5T HDx LPCA Port A 14x sw	3.0T HD LPCA Legacy Port 12x sw	Upgrade to 3.0T HDx LPCA Legacy Port 14x sw	3.0T HDx LPCA Port A 14x sw
6ch Flex PA	X	X	X	X	X	X
6ch Flex PA 1	X	X	X	X	X	X
6ch Flex PA 2	X	X	X	X	X	X
3ch Flex PA 1	N/A	N/A	X	N/A	N/A	X
3ch Flex PA 2	N/A	N/A	X	N/A	N/A	X

4. If the appropriate coil name is not in the list, search the **Add available GE coils** list in the upper-right corner and add the coils, if available.
5. If the coil is not available in any of the Coil Config lists, load the latest available software and software patches for your system according to the "Installing Software Patches and Service Packs procedure on the service procedures website, <http://3.87.118.28/optec3/common/hdx/content/495398.htm>.
 - To fully install the coil after loading a Service Pack, a reboot of the system is required.

Performance check

The coil performance check uses two different protocols for Signal and Noise image acquisition to check image quality. The Signal scan is a Fast Spin Echo (FSE) sequence used to minimize susceptibility and B0 inhomogeneity effects. The Noise scan is a Gradient Echo (GRE) sequence that has a Control Variable (do_noise) to eliminate the transmit radio frequency (RF) completely during the acquisition. The Signal scan must be run prior to the Noise scan as the same R1 and R2 values from the Signal scan are used in the noise scan.

An image quality check should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals. Table 3-2 lists the tools needed to check the quality of the coil imaging performance.

Table 3-2 Performance verification tools

Description	Part number	Quantity
1.5T Head Loader	46-287899G1	1
1.5T Head SNR TLT Sphere	46-265826G6	1
3.0T Head Loader	2360031	1
3.0T Head TLT Sphere	2359877	1
NOTE: The phantom part numbers listed in the table match forward production HDx systems. For HD and upgrades, the phantom part numbers for the Head Loader and Head TLT Sphere may be different. However, they are okay to use.		

NOTE: Positioning pads and straps as required. The patient restraint strap that attaches to the table works well to hold the upper coil half in place when it is positioned on top of the Head Loader phantom.

Signal scan

To acquire a Signal scan, perform the following steps:

1. From the Scan desktop, click **New Patient** to start a new scan.
2. Enter **geservice** in the **Patient ID** text box and press **Enter**.
3. Enter **111** pounds (50 kg) in the **Patient Weight** text box.
4. If the Patient Protocols panel for setting up a protocol is not visible, click **Patient Position** (located below the Protocols text box) to open the protocols panel.
5. Place one of the coil halves in the posterior position under the **Head Loader Phantom** on the patient table (Figure 3-4).
6. Position the **Head SNR Sphere** inside the **Head Loader**.
7. Position the other coil half and plug in the **Quick Disconnect (QD) Box**.
 - a) Place the other coil half on top of the **Head Loader** and secure the two coil halves together using the patient restraint strap attached to the table cradle.
 - b) Plug the upper coil half into Position 1 on the **QD Box** and plug the lower coil half into Position 2 on the **QD Box**.
 - c) Plug the **QD Box** into the **LPCA**. Make sure the coil cables are routed properly and not looped or crossing each other.

Figure 3-1 1.5T Legacy Phased Array Port C

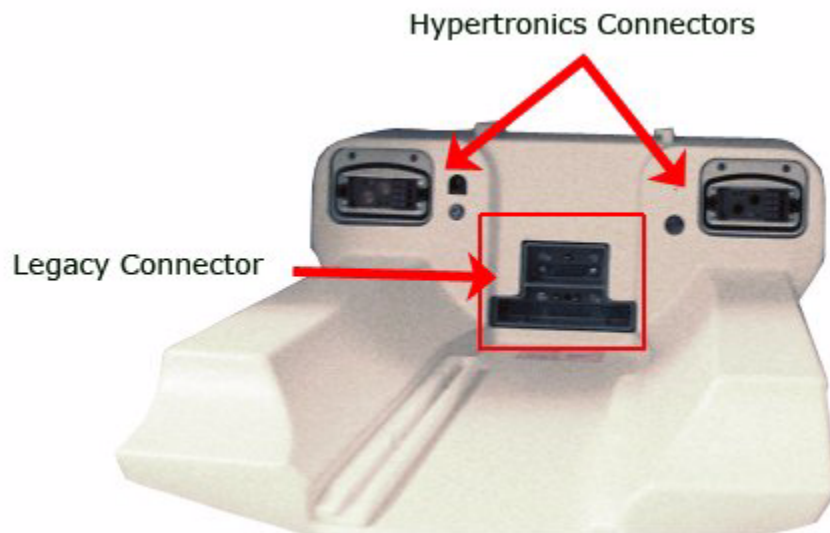


Figure 3-2 3.0T Legacy Phased Array Port C



Figure 3-3 1.5T/3.0T HDx Port A with System Preamps



Figure 3-4 Coil and phantom setup

Head PA
coil upper half

Head PA
coil lower half



TLT phantom
with loader

8. Set the landmark using the alignment lights to the cross hairs marked on the anterior half of the coil.
 - a) At the magnet, press the **Alignment Light** button to turn on the laser.
 - b) Move the cradle to align the coil landmarks to the alignment lights.
 - c) Press the **Landmark** button to landmark the alignment.
9. Move the coil to scan position by pushing the **Advance to Scan** button, ensuring that the cable does not get pinched or ensnared between the tube/bore and the moving patient cradle.
10. At the operator console, set up the protocol for the Signal scan. See [Table 3-3](#) for 1.5T systems and [Table 3-4](#) for 3.0T systems. The tables are located after the Noise scan procedure.
11. Click **Save Series** to download the protocols.

12. Click **Download**.
13. Adjust the Control Variables to save the intermediate images from a scan to facilitate ROI measurements.
 - a) Right-click the **Research Options** tab and select **Display CVs**.
 - b) In the CV Name text box, type **saveinter** and press **Enter**. In the Current Value text box, type **1** and press **Enter**.
 - c) Click **Accept**.
14. Click **Auto Prescan**.
15. Record the R1, R2, TG, and System Frequency values from the results of the Auto Prescan on the [SNR data sheet](#) in Appendix A.
16. Click **Scan**.
17. After the scan is complete, continue to the next section and acquire a [Noise scan](#).

Noise scan

A [Signal scan](#) must be run prior to the noise scan as the same R1 and R2 values must be used for both the Signal and Noise scans. Do not run an Auto Prescan prior to the Noise scan, as the values will change.

To acquire a Noise scan, perform the following steps:

1. In the Rx Manager, copy the Signal scan series.
 - a) Right-click the series you want to copy select **Copy Series**.
 - b) Right-click in the Rx Manager and select **Paste Series**.
 2. Click **View Edit** and set up the protocol for the Noise scan. Refer to [Table 3-3](#) for 1.5T systems and [Table 3-4](#) for 3.0T systems. The tables are located after the Noise scan procedure.
 3. Click **Save Series**.
 4. Click **Download**.
 5. Adjust the research Control Variables.
 - a) Right-click the **Research Options** tab and select **Display CVs**.
 - b) In the CV Name text box, type **saveinter** and press **Enter**. In the Current Value text box, type **1** and press **Enter**.
 - This saves the intermediate images from a scan to facilitate ROI measurements.
 - c) In the CV Name text box, type **rhformat** and press **Enter**. In the Current Value text box, type **1** and press **Enter**.
 - This sets grad warp to off.
 - d) In the CV Name text box, type **do_noise** and press **Enter**. In the Current Value text box, type **1** and press **Enter**.
 - This eliminates the transmit RF completely during the acquisition.
 - e) Click **Accept**.
 6. Click **Manual Prescan**.
- NOTE:** Do NOT make any changes to the parameters in Manual Prescan because the R1 and R2 values must remain the same as the Signal scan values. The TG value could change, but no adjustments are to be made.
7. Click **Done**.
 8. Click **Scan**.
 9. After the scan is complete, continue to the [SNR image analysis](#) section.

Signal and Noise protocols

Table 3-3 1.5T Signal and Noise protocols

1.5T Protocols		
Parameters	Signal scan entries	Noise scan entries
Patient/Exam information		
Patient ID	geservice	geservice
Patient Name	6ch Flex PA QA	6ch Flex PA QA
Patient Weight	111 lbs (50 kg)	111 lbs (50 kg)
Patient Position		
Patient Position	Supine	Supine
Patient Entry	Head first	Head first
Coil	6ch Flex PA	6ch Flex PA
Series Description	Signal	Noise
Imaging Parameters		
Plane/Mode	Axial /2D	Axial /2D
Pulse Sequence	FSE-XL	GRE
Grad Mode (TRM coil systems only)	Whole	Whole
Imaging Options	Fast	None
Scan Timing		
# of Echoes	1	1
TE/TR	17/500	MinFull/34
Echo Train/Flip Angle	4	1
Bandwidth	15.63	15.63
Acquisition Timing		
Freq/Phase	256/256	256/256
NEX	1	1
Phase FOV	1	1
Freq DIR	A/P	A/P
Shim	Auto	Auto
Phase Correct	On	Not selectable
Scanning Range		
FOV	38	38
Slice Thickness/Spacing	3/1.5	3/1.5
Start, End, R/L Center, P/A Center, Table Delta	0, 0, 0, 0, 0	0, 0, 0, 0, 0
Slices	1	1

Table 3-4 3.0T Signal and Noise protocols

3.0T Protocols		
Parameters	Signal scan entries	Noise scan entries
Patient/Exam information		
Patient ID	geservice	geservice
Patient Name	6ch Flex PA QA	6ch Flex PA QA
Patient Weight	111 lbs (50 kg)	111 lbs (50 kg)
Patient Position		
Patient Position	Supine	Supine
Patient Entry	Head first	Head first
Coil	6ch Flex PA	6ch Flex PA
Series Description	Signal	Noise
Imaging Parameters		
Plane	Axial	Axial
Mode	2D	2D
Pulse Sequence	FSE-XL	GRE
Grad Mode (TRM coil systems only)	Whole	Whole
Imaging Options	Fast	None
Scan Timing		
# of Echoes	1	1
TE/TR	17/1,000	MinFull/34
Echo Train/Flip Angle	4	1
Bandwidth	15.63	15.63
Acquisition Timing		
Freq/Phase	256/256	256/256
NEX	1	1
Phase FOV	1	1
Freq DIR	A/P	A/P
Shim	Auto	Auto
Phase Correct	On	Not selectable
Scanning Range		
FOV	38	38
Slice Thickness/Spacing	3/1.5	3/1.5
Start, End, R/L Center, P/A Center, Table Delta	0, 0, 0, 0, 0	0, 0, 0, 0, 0
Slices	1	1

SNR image analysis

An ROI should be used to measure the signal and noise on your acquired coil quality images. Regions of interest in both signal and noise images can be measured directly in the Viewer.

The SNR is calculated using the signal-to-noise ratio of the composite image:

$$SNR_i = \frac{\text{mean of signal within ROI}}{\text{standard deviation of noise within ROI}}$$

$$SNR_C = \sqrt{\sum_{i=1}^2 SNR_i^2}$$

NOTE: The SNR calculation uses the mean of the signal image and standard deviation of the noise image.

SNR measurement

To acquire measurements, perform the following steps for the Signal ([Figure 3-5](#)) and Noise ([Figure 3-6](#)) scans:

1. Select the Signal scan, image 7, in the Browser and click **Viewer**.
2. In the Viewer control panel, click **Format** and select the single image format.
 - Make sure to use image 7, the composite image, from the signal or noise series for the ROI measurements.
3. Review the image for any artifacts, i.e., zippers, sever ghosting, or corduroy. If an artifact is found, correct the issue before continuing, as this can impact the coil SNR results.
4. In the Viewer control panel, click **Measure** and select the **Circular ROI** icon.
5. Click and drag to adjust the ROI size and orientation.
 - Choose a circular ROI of ~16,000 mm² covering the appropriate section of the phantom.
 - Mean, standard deviation, and area of the ROI will appear in the lower-right corner of the image.
6. Record the Signal Mean value and the Area value on the [SNR data sheet](#).
7. Make sure the ROI display is highlighted (blue in color). If not, select it and simultaneously press **Ctrl** and **C** to copy the ROI.
8. Select the Noise scan, image 7, in the Browser and click **Viewer**.
9. Simultaneously press **Ctrl** and **V** to paste the ROI.
10. Record the Noise Standard Deviation value on the SNR data sheet.
11. Perform the following SNR calculation:

SNR = Signal Mean Value (ROI) / Noise Standard Deviation Value (ROI)

 - To use the calculator application on the scanner, right-click in the clear area at the bottom of the Service Tool desktop and select **Calculator**.
12. Record SNR results in the SNR data sheet and compare to specifications. If the results are below the required specifications, refer to the [Troubleshooting](#) section of this manual or escalate the issue.

Figure 1: A plot of the intensity distribution of the beam at the center of the detector. The plot shows a semi-circular distribution of intensity on a grid. The x-axis is labeled 'x' and the y-axis is labeled 'y'. The intensity is highest at the center and decreases towards the edges. The plot is titled 'FIELDVIEW' and includes a date '20 Apr 2006' and a page number 'Page = 1/3'.

[illegible]

Figure 9: A plot of the field of view (FOV) of the camera. The plot shows a grid with a central bright spot and a surrounding dark area. The grid is labeled with 'x' and 'y' axes. The central spot is labeled 'FOV' and '1000'. The surrounding area is labeled '1000' and '1000'. The plot is titled 'Figure 9' and 'FOV of the camera'.

Figure 3-10 Receiver 4

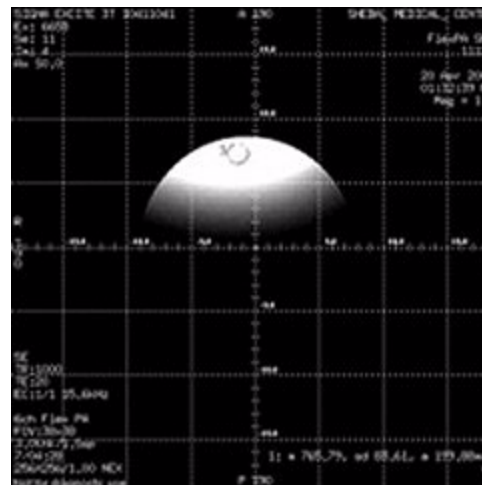


Figure 3-11 Receiver 5

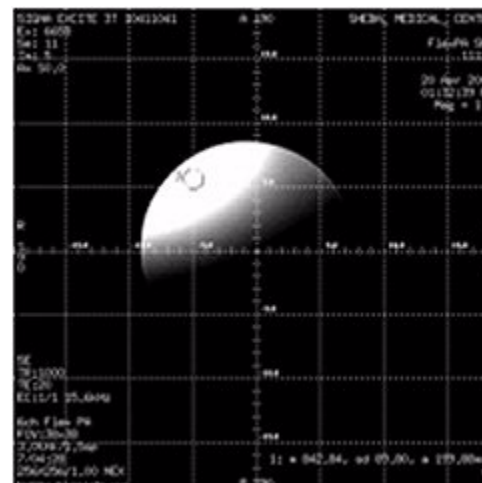
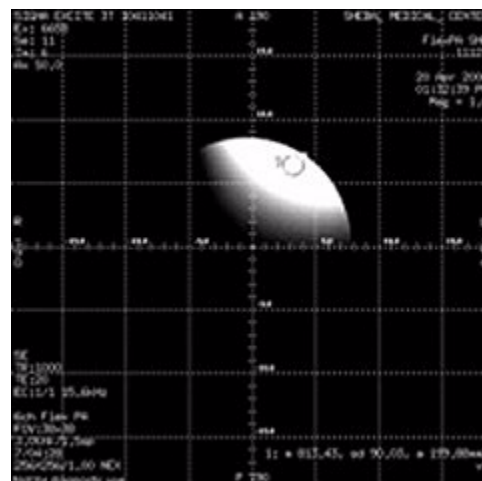


Figure 3-12 Receiver 6



Composite image SNR specification

The SNR measurements must be greater than or equal to the specifications in Table 3-5.

Table 3-5 SNR specifications

Coil	Composite image
1.5T 6-ch Flex PA	240
3.0T 6-ch Flex PA	230

Chapter 4

Maintenance

Introduction

Maintaining a controlled environment involves routine cleaning and maintenance checks of your **6-Channel Flexible Phased Array coil**. Careful planning and diligent upkeep of your coil can provide a safe environment for both patients and employees.

Be aware of the cleaning guidelines and required maintenance for the following:

- [Coil and components care](#)
- [Cleaning](#)
- [Storage](#)

Coil and components care



WARNING: Detach the coil and intermediate connector from the scanner before attempting to clean. Do not reattach after cleaning until the coil and intermediate connector have dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

Cleaning

The coil, intermediate connector, and patient comfort pads must be cleaned and stored after each use using the following procedures:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- Do not spray or pour any cleaning solution directly on the coil or the intermediate connector!
- Let the coil, intermediate connector housing, and pads dry before use.
- Store the coil and the intermediate connector in an air-conditioned scan room or equipment room.

IMPORTANT: Under no circumstances should the coil or intermediate connector be placed into any type of sterilizer!

Storage

Store the coil and intermediate connector in an air-conditioned scan room or equipment room. To store the coil, a storage space of 50 cm (width) × 60 cm (depth) × 30 cm (height) is required.

Chapter 5

Troubleshooting / Spare parts

Introduction

The two coil halves of the **6-Channel Flexible Phased Array coil** assembly are identical in design and manufacture. The foam sealed housing for the coil provides the required flexibility and ease of use in a sterile operating room environment. This however restricts access to the internal components, so troubleshooting the coil is performed at the coil half assembly level. The coil halves are not matched pairs and do not require tuning so individual coil halves can be ordered and replaced.

Each coil half plugs into an Intermediate Adaptor Connector, otherwise known as a Quick Disconnect (QD) box, which is then plugged into the Low Profile Carriage Assembly (LPCA) on the scanner. No imbedded pre-amps are provided in the coil or the QD Box, pre-amplification is done inside the LPCA. See Table 5-1 for Coil ID information.

Table 5-1 Coil ID

Coil assembly	Coil ID	Coil ID hardware
1.5T 6-Channel Flex Phased Array coil assembly with a Legacy Intermediate Adaptor Connector - System Type: HD and upgrade to HDx - Mates to LPCA Legacy Port C	No	None A single col half can be used independently when the appropriate coil configuration name is selected from the Protocol menu.
1.5T 6-Channel Flex Phased Array coil assembly with a Hypertronic Intermediate Adaptor Connector - System Type: forward production HDx - Mates to LPCA Port A	Yes	Three Coil ID Chips located in the QD Box - one is used for the recognition of both coil halves being connected - one is used for the recognition of a coil half plugged into QD Box position 1 - one is used for the recognition of a coil half plugged into QD Box position 2 - this configuration allows you to scan with only one coil half connected
3.0T 6-Channel Flex Phased Array coil assembly with a Legacy Intermediate Adaptor Connector - System Type: HD and upgrade to HDx - Mates to LPCA Legacy Port C	Yes	One Coil ID Chip located in the QD Box - Requires both coil halves to be connected to the QD Box for coil recognition and system functionality - Failure to do so results in "Multicoil Bias Error"

Coil assembly	Coil ID	Coil ID hardware
3.0T 6-Channel Flex Phased Array coil assembly with a Hypertronic Intermediate Adaptor Connector - System Type: forward production HDx - Mates to LPCA Port A	Yes	Three Coil ID Chips located in the QD Box - One is used for the recognition of both coil halves being connected - One is used for the recognition of a coil half plugged into QD Box position 1 - One is used for the recognition of a coil half plugged into QD Box position 2 - This configuration allows you to scan with only one coil half connected

You should be familiar with the following troubleshooting guidelines and spare parts for the **6-Channel Flexible Phased Array coil**:

- [Troubleshooting](#)
- [Spare parts](#)

NOTE: Other than replacing the spare parts listed in [Table 5-2](#), no other field service intervention is allowed.

Troubleshooting

See the [Introduction](#) section for a coil function overview, which will aid in troubleshooting coil or system problems.

Hardware inspection

There are a few quick inspections that can be done initially to help understand where the problem might lie. Perform these steps to troubleshoot a hardware problem.

1. Inspect the foam housing of each coil half and look for cracks, discoloration or other damage.
2. Inspect the cables and look for signs of damage such as cuts in the insulation, kinks or twists, especially at the coil and connector entry points.
3. Inspect the cable connector and connector pins. Look for signs of damage to the pins such as corrosion, bent pins, or missing pins.
4. Make sure the field strength labeled on the coil is correct for the scanner it is used on.
5. Inspect the Intermediate Adaptor connectors for bent or missing pins.
6. Inspect the Intermediate Adaptor housing for cracks, discoloration, or other damage.
7. Inspect the connectors on the LPCA and look for damaged or missing pins.
8. Make sure the coil cable connector gets fully engaged into the Intermediate Adaptor connector.
9. Make sure the Intermediate Adaptor gets fully engaged into the LPCA connector.

System inspection

Perform these steps to troubleshoot a system problem.

1. Verify the scanner's operating frequency is within the bandwidth of the coil. See the [Requirements and specifications](#) section in Chapter 2 of this manual and look for the bandwidth specification for the coil. Corrections to system operating frequency may be necessary to improve coil performance.
2. Review the customer's images and look for shading or artifacts.

Troubleshooting tips

1. Perform an SNR test on the coil and compare the results to the baseline that was performed when the coil was put into service.
2. If the SNR is low or an open / short circuit fault occurs, review the Element Performance as defined in this user manual to determine which channels are having problems.
3. To determine if the problem lies in the coil or the receive chain, try swapping the coil cables at the interface adaptor to see if the problem moves with the cable swap. See [Figure 5-1](#). If the problem moves then the issue is with the coil half.
4. If the problem does not move with the cable swap, use the MCR tool to determine where in the receive chain the problem lies.

Figure 5-1 Coil cables swapped at interface adaptor



Spare parts

Table 5-2 describes the spare parts for the **6-Channel Flexible Phased Array coil**.

Table 5-2 6-Channel Flex coil spare parts

Description	Part number
1.5T 6-Channel Flex Phased Array coil half (each coil half is identical; therefore, a single half is permitted for replacement)	5229623
3.0T 6-Channel Flex Phased Array coil half (each coil half is identical; therefore, a single half is permitted for replacement)	5229646
1.5T Legacy Intermediate Adaptor Connector	5193663
1.5T Hypertronics Intermediate Adaptor Connector	5199090
3.0T Legacy Intermediate Adaptor Connector	5260688
3.0T Hypertronics Intermediate Adaptor Connector	5260689
6-Channel Flex Coil Accessories Spares Kit (needed only when the coil is used on the Surgical Suite product; this kit contains replacement parts to support the coil strain relief and mounting mechanism)	5269901
Operator/Service manual (English)	5267963-2EN
Operator/Service manual (Multi-language)	5267963-299

IMPORTANT: A Functional SNR test should be completed after replacing any of the hardware parts.

Appendix A

SNR data sheet

Introduction

This appendix contains the signal-to-noise ratio (SNR) data sheet. This is used during SNR image analysis to record the values for the mean, standard deviation, and SNR calculation. For detailed information on SNR image analysis and SNR measurement, refer to the [Quality assurance](#) chapter.

Table A-1 SNR data sheet

[illegible]

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